

- A. Please describe how the Unzip parallel pattern works. How is it implemented (pseudo code and PRAM complexity required)? (2)

[illegible]

- B. How and under what conditions can the following code be parallelized? Is there a way to write the code for better parallelization? Assume that a and b store integers. (2)

```
for (i=4; i<100; i++) {
    for (j=1; j<100; j++)
        a[i][0] *= f(a[i][j]);
    for (j=0; j<100; j++)
        b[i-4][j] += f(b[i][j]);
}
```

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- A. Does it make sense to use the corner-turning optimization in a non-GPU-based architecture? Why? (1) **This question can be skipped if you got 1 or 3 points in the second challenge.**

- B. Please briefly describe the differences between Memory coherence and memory consistency. (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**

- C. Please describe how merge scatter works. (1)

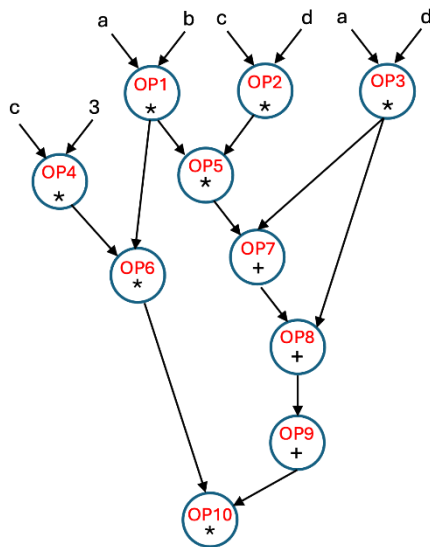
- A. Is an execution configuration of $\langle\langle\langle 120, 120 \rangle\rangle\rangle$ suitable for a GPU with 40 streaming processors and 32-thread warps? (1)

- B. Considering the execution configuration of the previous question, how many copies of a shared variable will exist during the execution of the kernel? (1)

- C. What is on-demand migration and why is it harmful for performance? (1)

- D. Control divergence only happens when the kernel contains if-else blocks. Is this statement true or false? Explain. (1)

- A. Consider the following DFG and HLS schedule obtained with the list-based algorithm. What can you infer about the constraints on resources? (2)



Clock cycle	Scheduled operations
1	OP1, OP2
2	OP5, OP3
3	OP7, OP4
4	OP8, OP6
5	OP9
6	OP10

- B. How does the schedule from the previous question change if the available resources are 4 multipliers and 2 adders (all with 1 clock cycle latency)? (1)

- C. When is cudaSetDevice() used? (1)

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